Anti-Forensics with Steganographic Data Embedding in Digital Images

Hung-Min Sun, Chi-Yao Weng, Chin-Feng Lee, and Cheng-Hsing Yang

***Abstract*—E-forensics investigates and extracts confidential information from electronic products; in other words, the anti-forensics indicates that evidences in those products are imperceptible and undetected. This work presents an anti-forensic steganography method that can embed and extract messages from images. Highlight of exploiting modification direction (HoEMD) and adaptive EMD (AdEMD) methods use the module operation and take into account of the sensitive nature of a human visual system. The HoEMD approach exploits the pixel directions. A pixel with a larger change implies more pixel directions and, ultimately, a larger embedding capacity. The pixel differencing in the proposed AdEMD method is used to evaluate whether the pixel located in the edge area can tolerate a larger change than that of the pixel location in a smooth area. To successfully extract a message, the differencing value is maintained on the same level before and after data is concealed; a delicate adjusting phase is used as well. In contrast with the PVD and LSB replacement method of Wu et al., LSB substitution method, and Lee and Chen's data hiding scheme based on modulus function, the proposed steganography system has a larger embedding capacity and a higher image quality. Effectiveness of the proposed steganography schemes over that of a previous blind steganalyzer is demonstrated using the statistical attack of Chi-square analysis.**

***Index Terms*—e-forensic, anti-forensics, human visual system (HVS), exploits pixel directions, steganalysis.**

I. Introduction

DIGITAL forensics detects and observes abnormal conditions in digital applications or devices, including computers, PDAs, and networks, to determine the evidence [1], [2]. In particular, Healy has extensively studied the feasibility of applying E-forensics in information science to digital devices such as mobile phones[3]. E-forensics generally refers to identifying, observing, preserving, and analyzing digital evidences by complying with standard operation procedures. In 2006, Harris proposed a new consensus in anti-forensics and traced the problems in anti-forensics [4]. Further, Distefano et al. extended the issues of anti-forensics and applied anti-forensics techniques to mobile devices[5]. Nevertheless, the anti-forensics closely resembles data concealment owing to the same target that hides more evidences or messages into devices against detection and imperceptibility.

As a scientific method, steganography hides data for various purposes, including security protection and authentication. Steganography embeds messages into meaningful images, referred to as cover images, without causing attention to malicious individuals. Data hiding focuses on large embedding capacity and high image quality after messages are hidden in cover images. However, large embedding capacity and high image quality create a trade-off situation, since a large embedding capacity always incurs a significant amount of image distortion. Therefore, it necessitates a closer examination of the balance between high embedding capacity and high image quality. Considerable efforts have focused on high embedding capacity or high image quality in data hiding [6]–[16]. A simple and conventionally adopted steganography method, least-significant-bit substitution (LSB), embeds k secret bits into a cover pixel by replacing k LSBs [6], [7]. Despite performing better than other approaches [9]–[12], the LSB approach is easily detected by some malicious programs[19]–[21] and does not consider human vision[13]–[16]. The feasibility of achieving a high embedding capacity from a human vision aspect has received considerable attention.

To achieve high embedding capacity, Wu and Tsai developed a pixel-value differencing (PVD) method[13], in which the pixel difference between two neighboring pixels are used to estimate how many bits will be embedded. Jung et al. developed a multi-pixel differencing and LSB substitution approach, where a block with four pixels is used to embed data in the multi-pixel differencing way[14]. Lee and Chen proposed a data hiding scheme based on the modulus function, in which a variant Cartesian product is operated upon the modulus function to hide the message [15]. The above approaches focus on achieving a high embedding capacity.

While attempting to achieve high image quality [9]–[12], Wang *et al.* improved the image quality by using the modulus function to decrease the discrepancy that is caused by utilizing the PVD approach [12]. Zhang and Wang devised a high quality data hiding strategy based on exploiting modification direction (EMD) [11]. Despite achieving high quality, the above approaches have seldom considered embedding capacity.

To achieve a high performance approach, which contains both large embedding ratio and high image quality, is an important issue [18]. This paper presents a high performance approach using the adaptive EMD method. Experimental results demonstrated that the proposed approach performs

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better than those in literature in terms of embedding capacity and image quality. For security, the proposed approach is more secure than previous ones in terms of passing the blind steganalyzers, as evidenced by Chi-square testing [20].

II. Literature Review

This section describes two high performance methods with anti-forensics steganography, i.e. data hiding scheme based on the modulus function (HBMF for short) by Lee and Chen and exploiting modification direction (EMD for short) developed by Zhang and Wang. The HBMF focuses on the branch area, high capacity, of data hiding by using the modulus function. In their method, the size of the variant Cartesian product is examined to evaluate the feasibility of embedding more confidential messages. The EMD method contributes to high quality since the cover pixel is slightly modified, indicating that cover pixels are changed among [-1, 1], for the embedding of confidential messages. These two approaches are described as follows.

A. Lee and Chens data hiding scheme based on the modulus function (HBMF)

Lee and Chen [15] developed the HBMF approach, in which the variant Cartesian product upon the modulus function is applied to hide and extract the message. In their manner, first of all, two set-hash functions H_r and H_l are used to generate two sets of K_r and K_l, containing 2^α and 2^β elements, respectively. Then the Cartesian product is applied to generate an ordered set of the variant Cartesian product with $2^\alpha \times 2^\beta$ elements. The binary stream is divided into secret piece s_k with $(\alpha+\beta)$-bit. Then, the secret piece s_k is hidden in a pixel group G_p. The basic rules are as follows: Let the pixel intensity set PG be an ordered set of pixel values that dominates the pixel values of a cover image, i.e. pixel intensity set with 8-bit gray-level is $PG = \{x_i | 0 \leq x_i \leq 255; i = 1$ to the total number of pixels in images$\}$. The pixel group G_p is a subset of pixel intensity set PG and is created by $r = x_i \bmod n$, where n is a size of the variant Cartesian product ($=2^\alpha \times 2^\beta = 2^{\alpha+\beta}$). As a result, the pixel group G_p is generated and represented as an ordered set of $\{x_i - r, x_i - r + 1, ..., x_i, x_i + 1, ..., x_i + n - r - 1\}$.

B. Zhang and Wang's EMD

Zhang and Wang developed the exploiting modification direction (EMD) scheme in 2006. In this scheme, the secret S must first be translated into a $(2n+1)$-ary notation, denoted as $(S_1, S_2, S_3, ..., S_m)_{2n+1}$. Each secret S_i is carried by a pixel group with n cover pixels $(g_1, g_2, ..., g_n)$, where length l of S_i is determined by $l = \lfloor log_2(2n+1) \rfloor$. For each pixel group, the critical function f is calculated, which is defined as $f(g_1, g_2, ..., g_n) = [\sum_{i=1}^{n}(i \times g_i)] \bmod (2n+1)$. If the result of the critical function is equivalent to the to-be-embedded secret S_i, all of the pixels in a pixel-group remain unchanged. Otherwise, only one pixel in a pixel-group $(g_1, g_2, ..., g_n)$ is increased or decreased by one after S_i is hidden. For the extracting procedure, the critical function f is calculated by the pixel group and the result equals the secret S_i.

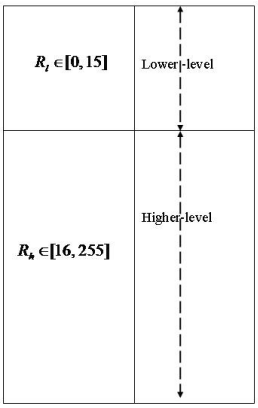

Fig. 1. An example of l-h division table with division line D_{lh} = 15.

III. Our Proposed Approach

This work presents two novel approaches, the highlight of EMD (HoEMD) and the adaptive EMD (AdEMD), to achieve high efficiency, high quality and large embedding ratios. The proposed schemes are based on modified pixel directions while the HoEMD approach considers all pixel directions. A general critical function f is derived for the embedding procedure of HoEMD. More importantly, achieving a high embedding performance depends on the fluctuating pixel value. The proposed approach can gain a high embedding capacity since setting the difference in value of k is larger. For instance, the pixel value is modified by k, then the embedding ratio (ER) is $ER = \frac{EC}{n} = \frac{\lfloor \log -2(R)^n \rfloor}{n}$, where n denotes the pixel-group size. The proposed adaptive EMD (AdEMD) approach incorporates the use of a human visual system (HVS), in which smooth/flat areas have a lower tolerance of a significant change in the pixels than in the edge areas. Thus, the pixel difference belonging to smooth or edge areas is evaluated using a quantization table (shown in Fig. 1), which is divided into two difference levels, a lower one and a higher one. The lower level refers to a situation in which the pixels are located in smooth areas, thus preventing a significant amount of distortion. Conversely, a higher level refers to a situation in which the pixels are located in edge areas, allowing them to tolerate more significant changes. To successfully extract the secret messages, the pixel differences must fall into the same level before and after data are embedded. If the difference value shifts to another level after data are embedded, the pixel values are readjusted using an adjusting phase. The following subsections describe the details of the embedding and extracting algorithms of the proposed approaches.

A. HoEMD

For simplicity of description, we specified some symbols to describe our approach. The notation is defined as follows.

k: The largest changed value of a pixel
EC: Embedding capacity
Ψ: The number of total pixel directions
f: Critical Function
B: Binary stream
S: Secret digitals
n: pixel-group with n pixels

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The value of k in the proposed embedding procedure is determined first. A pixel-group with n pixels is selected by users or a random generation. Notably, the embedding ratio is a fixed value (ER=$\frac{EC}{n} = \frac{\lfloor \log_2(\Psi)^n \rfloor}{n}$), where Ψ=2k+1. Thus, we recommend that n is 2 for a lower computation complexity. Length l of binary stream B is then instructed to transform into a sequence of secret digits S, where length l of binary stream B is $l = \lfloor \log_2(\Psi)^n \rfloor$. The form of secret digits S is $(S_1, S_2, S_3, \cdots, S_n)_{\Psi^n}$. Next, a critical function f in data embedding and extracting procedures is defined.

Let a secret digit S_j be the digit to be embedded into a pixel-group. The critical function f,$f(g_1, g_2, ..., g_n) = [\sum_{i=1}^{n}(\Psi^{i-1} \times g_i)]$ mod Ψ^n is applied, and the difference value d between f and S_j is computed. If d = S_j-f = 0, all n pixels in the pixel-group remain unchanged. However, if d=(S_j-f) $\neq 0$, the new difference d' is calculated by applying the shifting strategies. The transformed n-dimensional vector is then obtained from the new difference d', in which the n-dimensional vector is based on the Ψ notation system. Finally, the result of the pixel group in the stego-image operates two vectors of the pixel group $(g_1, g_2, \cdots, g_{n-1}, g_n)$ and n-dimensional vector $(R'_1, R'_2, \cdots, R'_{n-1}, R'_n)$, where the n-dimensional vector is the result of a variance of each pixel. The embedding-procedure is shown as follows.

1) Embedding-Procedure:

Input: A Cover image, S, n, k.

Output: A Stego-image.

Step 1: Divide a cover image into non-overlapping blocks with n pixels, where a block refers to a pixel-group containing n pixels.

Step 2: Derive the value of critical function $f(g_1, g_2, ..., g_n) = [\sum_{i=1}^{n}(\Psi^{i-1} \times g_i)]$ mod Ψ^n for each pixel-group, where each gray-value g_i is in [0, 255] and Ψ=2k+1.

Step 3: Estimate the difference value d between critical function f and the to-be-embedded secret digit S_j, where f denotes $f(g_1, g_2, ..., g_n) = [\sum_{i=1}^{n}(\Psi^{i-1} \times g_i)]$ mod Ψ^n. If d =S_j-f= 0, all pixels in the pixel-group remain unchanged. Otherwise, execute the following steps.

Step 3.1: Verify whether d falls into the range of $[\pm \left\lfloor \frac{(\Psi)^n}{2} \right\rfloor]$. If d falls out of the range, then d' is computed by Eq. (1) or Eq. (2). Otherwise, maintain the value as d'=d.

$$If\ d > \left\lfloor \frac{(\Psi)^n}{2} \right\rfloor, then d' = d - (\Psi)^n \quad (1)$$

$$If\ d < -\left\lfloor \frac{(\Psi)^n}{2} \right\rfloor, then d' = d + (\Psi)^n \quad (2)$$

Step 3.2: Comput $d'' = d' + \left\lfloor \frac{(\Psi)^n}{2} \right\rfloor$.

Step 3.3: Represent d'' as transformed n-dimensional vector based on Ψ, i.e. $d'' = (R_1, R_2, \cdots, R_n)_\Psi$, for each $R_i \in [0, \Psi - 1]$.

Step 3.4: Obtain a transformed n-dimensional vector $(R'_1, R'_2, \cdots, R'_n)$ based on the vector operation as $(R'_1, R'_2, \cdots, R'_n)$=$(R_1, R_2, \cdots, R_n)$+$(k, k, \cdots, k)$, where $(k, k, \cdots, k) = \left\lfloor \frac{(\Psi)^n}{2} \right\rfloor$.

Step 3.5: Calculate $\hat{G} = (g_1, g_2, \cdots, g_{n-1}, g_n)$ from adding two vectors $(g_1, g_2, \cdots, g_{n-1}, g_n)$ and $(R'_1, R'_2, \cdots, R'_n)$.

2) Extracting-Procedure:

Also used in the extraction-procedure, the critical function f is simple and easily implemented with its low computation complexity. The extraction procedure for each secret digit S_j is described below.

Input: Stego-image, n, k.

Binary bit-stream B

Step 1: Divide the Stego-image into non-overlapping blocks with n pixels, where a block is referred to as a pixel-group.

Step 2: Gain $\hat{S}$ by calculating the critical function f for each pixel-group.

Step 3: Transfer secret $\hat{S}$ into an l-bit string, where $l = \lfloor log_2(\Psi)^n \rfloor$

B. Adaptive EMD (AdEMD)

The gray-level image is used as the cover image in the proposed approach. First, the cover image is divided into non-overlapping blocks with two consecutive pixels, i.e. g_i and g_{i+1}. The difference value $diff$ is determined from each block. Before the data are embedded, the quantization table, as depicted in Fig. 1, is divided into lower and higher levels with dividing line D_{lh} = 15. Ranging from 0 to 255, the quantization table is divided into ranges R_l = [0, 15] and R_h = [16, 255], where the lower level contains R_l and the higher level contains R_h. The width of range R is denoted as $|R|$. The largest changed values of pixels (g_i, g_{i+1}) are denoted as $< k_1, k_2 >$. When the difference of the block falls into the lower level, the largest changed values of pixels (g_i, g_{i+1}) are set as $< lk_1, lk_2 >$. In contrast, if difference falls into the higher level, the largest changed values of pixels (g_i, g_{i+1}) are set as $< hk_1, hk_2 >$. Since the edge areas can permit a more significant change, the largest changed values for the higher level should be larger than that for the lower level. That is, the setting of the quantization table should be $lk_1, lk_2 \leq hk_1, hk_2$. Moreover, to successfully implement the adjustment phase, the restriction $lk_1 + lk_2 \leq |R_l|$ is used to limit the range of l-h division. Let S_j denote the digit to be embedded into a pixel-group. The value of S_j ranges from 0 to $\Psi_1 \times \Psi_2 - 1$, where $\Psi_1 = 2k_1 + 1$ and $\Psi_2 = 2k_2 + 1$. Detailed embedding steps of each block for the l-h division are as follows.

1) Data Embedding:

Step 1: Estimate the difference value $diff_i$ from each block with two pixels, g_i and g_{i+1}, i.e. $diff_i = |g_i - g_{i+1}|$.

Step 2: Identify the level which $diff_i$ belongs to in the l-h division. Let $< k_1, k_2 >$=$< lk_1, lk_2 >$ and $< hk_1, hk_2 >$, if $diff_i$ falls to lower and higher levels, respectively. Then, k_1 and k_2 refer to a situation in which the cover pixels of g_i and g_{i+1} are changed within $\pm k_1$ and $\pm k_2$, respectively.

Step 3: Apply the new critical function CF for each to-be-embedded secret digit S_j. The new critical function is defined as $CF = (g_i, g_{i+1}) = (g_i \text{ mod } \Psi_1)$+$(g_{i+1} \text{ mod } \Psi_2) \times \Psi_1$, where $\Psi_1 = 2k_1 + 1$ and $\Psi_2 = 2k_2 + 1$.

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Step 3.1: Determine the difference value d between critical function CF and secret digit S_j.

Step 3.2: If $d = S_j - CF = 0$, all pixels of the pixel-group remain unchanged; then, skip the algorithm and run the next pixel-group. Otherwise, execute the following steps.

Step 3.2.1: Transfer secret digital S_j int a 2-dimensional vector (R_1, R_2), where $R_1 = s \bmod \Psi_1$, $R_2 = \left\lfloor \frac{S}{\Psi_1} \right\rfloor$, $0 \le R_1 \le \Psi_1 - 1$, and $0 \le R_2 \le \Psi_2 - 1$. Then, calculate the value of V_1 and V_2 based on (g_i mod Ψ_1) and (g_{i+1} mod Ψ_2), where $(V_1, V_2) = ((g_i \bmod \Psi_1), (g_{i+1} \bmod \Psi_2)$.

Step 3.2.2: Calculate vector (R'_1, R'_2) by the vector operation as $(R'_1, R'_2) = (R_1, R_2) - (V_1, V_2)$, and apply the Eq. (3) to adjust the value.

$$R''_i = \begin{cases} R'_i - \Psi_i, & \text{if } R'_i > k_i \\ R'_i + \Psi_i, & \text{if } R'_i < -k_i \\ R'_i, & \text{otherwise} \end{cases}, \forall i = 1, 2. \quad (3)$$

Step 3.2.3: Obtain stego-pixel (g'_i, g'_{i+1}) from adding two vector (g_i, g_{i+1}) and (R''_i, R''_2).

Step 4: Determine new difference value $diff'_i$ by $diff'_i = |g'_i - g'_{i+1}|$

Step 5: If $diff_i$ and $diff'_i$ fall into different levels, execute the readjusting strategy.

Case 5.1: $diff'_i \in$ lower-level, readjust (g'_i, g'_{i+1}) to become the better choice of the following cases: $(g'_i - \Psi_1, g'_{i+1} - \Psi_2)$, $(g'_i + \Psi_1, g'_{i+1} - \Psi_2)$, $(g'_i - \Psi_1, g'_{i+1} + \Psi_2)$, $(g'_i - \Psi_1, g'_{i+1})$, $(g'_i, g'_{i+1} - \Psi_2)$, $(g'_i + \Psi_1, g'_{i+1}\Psi_2)$, $(g'_i + \Psi_1, g'_{i+1}$, $(g'_i, g'_{i+1} + \Psi_2)$.

Case 5.2: $diff'_i \in$ high-level, readjust (g'_i, g'_{i+1}) to become the better choice of the following cases: $(g'_i - \Psi_1, g'_{i+1} - \Psi_2)$, $(g'_i + \Psi_1, g'_{i+1} - \Psi_2)$, $(g'_i - \Psi_1, g'_{i+1} + \Psi_2)$, $(g'_i - \Psi_1, g'_{i+1})$, $(g'_i, g'_{i+1} - \Psi_2)$, $(g'_i + \Psi_1, g'_{i+1}\Psi_2)$, $(g'_i + \Psi_1, g'_{i+1}$, $(g'_i, g'_{i+1} + \Psi_2)$.

In Step 5, the better choice, i.e. (p'_i, p'_{i+1}), refers to a situation in which the pixel values satisfy that the difference values $d'_i = |p'_i - p'_{i+1}|$ and d_i fall into the same level, and $p'_i, p'_{i+1} \in [0, 255]$; in addition, the value of $(p'_i - p_i)^2 + (p'_{i+1} - p_{i+1})^2$ is smaller of those situations.

2) Data Extracting:

The data extracting phase resembles in the data embedding phase. First, the stego-image is partitioned into non-overlapping blocks with two consecutive pixels; the traversing order is also the same. The l-h division in the embedding procedure is reused here. For each block, the detailed extracting steps are described as follows.

Step 1: Estimate the difference value d'_i from each block with two pixels, i.e. g'_i and g'_{i+1}, and $d'_i = |g'_i - g'_{i+1}|$.

Step 2: From the l-h division, identify the level which d'_i belongs to. Let $< k_1, k_2 > = < l, l >$ and $< h, h >$ if d_i falls to lower and higher levels, respectively.

Step 3: Gain $\hat{S}$ by determining the new critical function CF from two pixels, g'_i and g'_{i+1}.

Step 4: Transfer secret $\hat{S}$ into an $length$-bit string, where $length = \lfloor log_2(\Psi_1 \times \Psi_2) \rfloor$

IV. Experimental Results and Analysis

Effectiveness of the proposed approaches, HoEMD and AdEMD, is tested for different types of images. Here, ten images with size 512×512 are used to perform the experiments. The secret binary string to be embedded is randomly generated by a Visual C++ program. For the HoEMD approach, n is equivalent to 2, and the largest changed values of the two pixels are k_1 and k_2. For the AdEMD approach, an l-h division and two embedding variables k_1 and k_2 are used. In terms of embedding capacity and image quality, performance of the proposed algorithms is evaluated by comparing our results with those of previous schemes.

A. Performance Analysis

The amount of data that can be concealed are evaluated using embedding capacity EC and embedding ratio ER (bit pre pixel), while the image quality after messages are embedded is evaluated using the peak signal to noise ratio (PSNR). To achieve a high embedding capacity, different values of k_1 and k_2 are used to demonstrate that the embedding capacity is increased since the values of k_1 and k_2 become larger. With larger values of k_1 and k_2, more accommodations are offered with a large number of directions because a large k_1 and k_2 generally have a high embedding capacity. Additionally, the embedding capacity EC of the proposed algorithm depends on the total pixel directions, but not all pixel directions can be used entirely. The embedding ratio ER is estimated as $ER = \frac{EC}{n} = \frac{\lfloor \log_2(R_1 \times R_2) \rfloor}{n}$, where $R_1 = 2k_1 + 1$ and $R_2 = 2k_2 + 1$, for setting various values of k_1 and k_2. If two setting values have the same secret accommodation, the setting value with smaller k_1 and k_2 values is preferred.

Tables I to IV list the qualities of the stego-images and the embedding performance ER setting at various $< k_1, k_2 >$ values. And arguments of $< k_1, k_2 >$ are set from 2 to 13 with $k_1 = k_2$ and $k_1 \neq k_2$ in our experimental result. Simulation resultants of value $< k_1, k_2 >$ with $k_1 < k_2$ is the same as value $< k_1, k_2 >$ with $k_1 > k_2$. For example, following the calculations of ER and PSNR, the performance of value $< 2, 3 >$ is equal to value $< 3, 2 >$. According to these results, the performance of ER is extremely high since the $< k_1, k_2 >$ values are larger. However, the setting value $< 6, 6 >$ in Table I has the same ER and lower image quality than setting value $< 5, 6 >$ in Table III.

In the HVS aspect, the edge areas allow a larger distortion than in the smooth areas. Therefore, an $l-h$ division table with division line D_{lh} = 15 is used to estimate a situation in which pixels fall into the lower level or the higher level. Table V summarizes the experimental results of the $l-h$ division table with division line D_{lh} = 15 for various k values. According to this table, the proposed approach correlates with the HVS concept in terms of providing a high embedding capacity and maintaining an acceptable image quality. Table VI compares the results of the proposed approach with those of the approach of Lee and Chen [15]. According to this table, our scheme indeed has a higher embedding ratio and image quality than that of[15]. On the average, the proposed approach with a pixel setting of $< 4, 4 > - < 7, 8 >$ obtains 10,486 more bits and a better 0.92 db image quality than that of [15]. For the

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case of $< 8, 8 > - < 11, 11 >$, the averaged evaluations in terms of capacity and image quality are 4.05 bpp and 33.54 dB, respectively, which are both higher than the compared scheme.

Table VII reveals that the proposed approach has a better embedding capacities and high image qualities[22] than that of the method of Wu *et al.* [16]; the image quality is the same as that of the simple 3-bit LSB approach. According to Table VII, the case of $< 8, 8 >$ in the proposed approach has the same average capacity of 4 bpp with a simple 4-bit LSB approach; the proposed approach achieves a better image quality, i.e. 2.97 db, than that of a simple 4-bit LSB approach.

B. Exception Procedures for Overflow/Underflow Probelems

Overflow/underflow problems occur since the cover pixel is near the boundaries 0 or 255. In the proposed embedding scheme HoEMD, the critical f is operated under the modulo number R^n. This finding implies that R^n difference varieties are accommodated. Additionally, the module operation has the characteristic of congruence numbers. Therefore, the pixel values must be adjusted so that the weight summation of the changed pixels is increased/decreased by R^n sequentially. For instance, assume that the $R = 7(= 2 \times 2 + 1)$, cover-pixels are $(254, 254)$, and secret digit $S_j = 19$. The HoEMD approach is applied to conceal the secret digit into cover-pixels, and the stego-pixels are $(257, 253)$. The overflow problem occurs since one value of stego-pixels is larger than 255. Therefore, the exception problem must be solved. The method involves subtracting R and $R - 1$ from two stego-pixels, so the stego-pixels are changed into $(250, 248) = (257, 253) - (7, 6)$. Notably, the weight summation of the changed pixels is $49(= 7 \times 1 + 6 \times 7)$, and $f = (250, 248) = (250 \times 1 + 248 \times 7)$ $\text{mod} 7^2 = 19$.

An example of the AdEMD approach is considered. Assume that $< k_1, k_2 >$ are $< 3, 4 >$, and secret digit $S_j = 30$. The critical function CF in the AdEMD approach is applied to embed the secret digit into two cover-pixels, and the stego-pixels are $(254, 256)$. The overflow problem occurs because one of the stego-pixels is more than 255. Thus, the stego-pixels with the overflow/underflow exception must be adjusted by $\pm R_i, i = 1, 2$. Therefore, the stego-pixels are changed into $(254, 247) = (254, 256) - (0, 9)$. The weight summation of the changed pixels is $30 = (254 \text{ mod} 7) + (247 \text{ mod} 9) \times 7$, which is equivalent to $f = (254, 256) = (254 \text{ mod} 7) + (256 \text{ mod} 9) \times 7 = 30$.

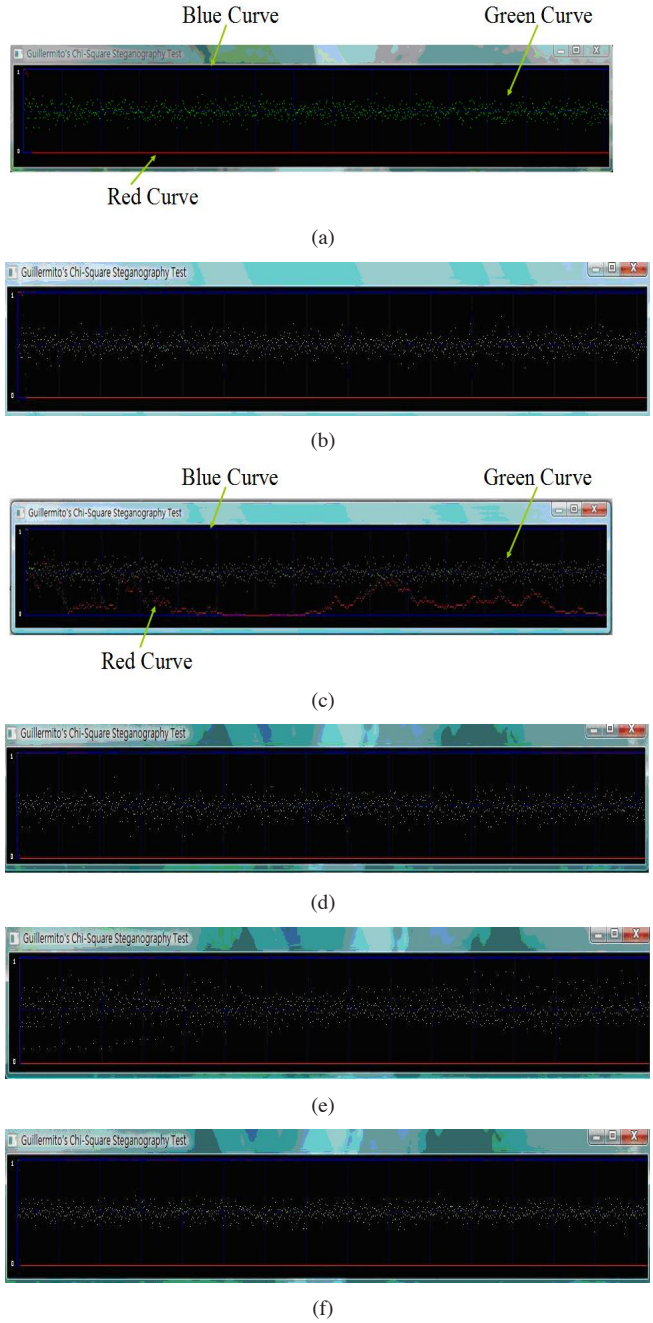

Fig. 2. The Chi-square-test results of image Lena for various approaches: (a)without using any approach; (b) EMD; (c) 1-bit LSB; (d) HoEMD with $< 2, 2 >$; (e) HoEMD with $< 7, 8 >$; (f) AdEMD with $< 4, 4 > < 11, 11 >$.

C. Security Analysis

Security is a vital concept for developing a data hiding scheme, thus a statistical steganalyzer of Chi-square testing is used to demonstrate that the proposed approaches can satisfy the security requirements. Three curves as displayed in Fig. 2, including the red one (the bottom line in the simulation result), the green one (distributed in middle of the simulation result), and the blue one (the top line in the simulation result), are depicted in the results of the Chi-square testing. In this analysis, the red curve indicates the outcome of the Chi-square test. In general, the red curve is reclined on the line of zero when a natural image is used as the test image; in other words the object has a high probability of embedding random messages if the red curve is close to one. Namely, secret data have been embedded in this object. The second output is the blue curve, in which each vertical blue line represents 1K bytes of embedded data. The third output is the green curve that remains around 0.5 if a random message is embedded. In Fig. 2, Figs. 2 (a), (b), (d), (e), and (f), are summarized as the undetected results since the red curves are close to zero. According to these results, the proposed approaches, HoEMD and AdEMD, can not be detected by using Chi-square testing as good as in a previous study of EMD.

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TABLE I
THE RESULTS OF EMBEDDING RATIO (ER) AND PSNRS USING OUR HOEMD APPROACH WITH VARIOUS $< k_1, k_2 >$ VALUES.

Cover image	$< 2,2 >$		$< 3,3 >$		$< 4,4 >$		$< 5,5 >$		$< 6,6 >$		$< 7,7 >$	
	ER	PSNR	ER	PSNR	ER	PSNR	ER	PSNR	ER	PSNR	ER	PSNR
Elaine	2	44.95	2.5	42.12	3	39.74	3	38.14	3.5	36.68	3.5	35.37
Lena	2	45.11	2.5	42.11	3	39.87	3	38.16	3.5	36.61	3.5	35.40
Baboon	2	45.13	2.5	42.11	3	39.86	3	38.13	3.5	36.62	3.5	35.41
Peppers	2	44.49	2.5	41.53	3	39.39	3	37.92	3.5	36.71	3.5	35.28
Toys	2	44.86	2.5	41.99	3	39.66	3	37.83	3.5	36.32	3.5	35.09
Girl	2	45.13	2.5	42.10	3	39.88	3	38.10	3.5	36.66	3.5	35.48
Gold	2	45.17	2.5	42.12	3	39.87	3	38.14	3.5	36.71	3.5	35.45
Barb	2	45.12	2.5	42.11	3	39.89	3	38.12	3.5	36.66	3.5	35.42
Zelda	2	45.11	2.5	42.11	3	39.89	3	38.15	3.5	36.69	3.5	35.41
Tiffany	2	45.18	2.5	42.01	3	39.87	3	38.13	3.5	36.53	3.5	35.45
Average	2	45.03	2.5	42.03	3	39.79	3	38.08	3.5	36.62	3.5	35.38

TABLE II
(CONT.)THE RESULTS OF EMBEDDING RATIO (ER) AND PSNRS USING OUR HOEMD APPROACH WITH VARIOUS $< k_1, k_2 >$ VALUES.

Cover image	$< 8,8 >$		$< 9,9 >$		$< 10,10 >$		$< 11,11 >$		$< 12,12 >$		$< 13,13 >$	
	ER	PSNR	ER	PSNR	ER	PSNR	ER	PSNR	ER	PSNR	ER	PSNR
Elaine	4	34.31	4	33.33	4	32.41	4.5	31.66	4.5	30.81	4.5	30.26
Lena	4	34.38	4	33.33	4	32.42	4.5	31.73	4.5	30.91	4.5	30.01
Baboon	4	34.39	4	33.69	4	32.15	4.5	31.64	4.5	30.64	4.5	30.26
Peppers	4	34.26	4	33.32	4	32.15	4.5	31.53	4.5	30.23	4.5	29.84
Toys	4	34.10	4	33.01	4	32.16	4.5	31.23	4.5	30.90	4.5	29.37
Girl	4	34.29	4	33.23	4	32.35	4.5	31.62	4.5	30.85	4.5	30.21
Gold	4	34.26	4	33.01	4	32.48	4.5	31.56	4.5	30.83	4.5	30.15
Barb	4	34.28	4	33.10	4	32.46	4.5	31.70	4.5	30.90	4.5	30.29
Zelda	4	34.41	4	33.31	4	32.47	4.5	31.73	4.5	30.94	4.5	30.24
Tiffany	4	34.24	4	33.23	4	32.37	4.5	31.71	4.5	30.80	4.5	30.10
Average	4	34.29	4	33.23	4	32.34	4.5	31.61	4.5	30.78	4.5	30.07

TABLE III
THE RESULTS OF EMBEDDING RATIO (ER) AND PSNRS USING OUR HOEMD APPROACH WITH VARIOUS $< k_1, k_2 >$ VALUES.

Cover image	$< 2,3 >$		$< 3,4 >$		$< 4,5 >$		$< 5,6 >$		$< 6,7 >$		$< 7,8 >$	
	ER	PSNR	ER	PSNR	ER	PSNR	ER	PSNR	ER	PSNR	ER	PSNR
Elaine	2.5	43.11	2.5	40.86	3	38.93	3.5	37.33	3.5	35.87	3.5	34.72
Lena	2.5	43.35	2.5	40.82	3	38.91	3.5	37.34	3.5	35.88	3.5	34.79
Baboon	2.5	43.40	2.5	40.86	3	38.90	3.5	37.32	3.5	35.86	3.5	34.72
Peppers	2.5	43.12	2.5	40.84	3	38.89	3.5	37.32	3.5	35.87	3.5	34.71
Toys	2.5	43.15	2.5	40.78	3	38.81	3.5	37.31	3.5	35.68	3.5	34.56
Girl	2.5	43.36	2.5	40.81	3	38.90	3.5	37.25	3.5	35.79	3.5	34.68
Gold	2.5	43.35	2.5	40.87	3	38.91	3.5	37.34	3.5	35.84	3.5	34.72
Barb	2.5	43.35	2.5	40.85	3	38.91	3.5	37.33	3.5	35.86	3.5	34.73
Zelda	2.5	43.36	2.5	40.83	3	38.87	3.5	37.32	3.5	35.87	3.5	34.72
Tiffany	2.5	43.37	2.5	40.83	3	38.90	3.5	37.31	3.5	35.86	3.5	34.63
Average	2.5	43.29	2.5	40.84	3	38.89	3.5	37.32	3.5	35.84	3.5	34.70

V. CONCLUSION AND FUTURE WORK

This work describes two novel anti-forensics approaches of steganogarphy, HoEMD and AdEMD, to hide data/messages into gray images while making the image imperceptive. The proposed HoEMD method can set the various changes, k values, of a pixel for high embedding capacity; in addition, the stego-image maintains an acceptable quality when the PSNR value exceeds 31 dB. The HVS concept is considered in the proposed AdEMD approach. Pixels located in the edge areas can tolerate larger values of hk_1 and hk_2 than those located in the smooth areas. Additionally, the quantization table is used to distinguish between the edge areas and smooth areas; the division line is defined by users. An exquisite adjustment phase is developed to ensure that the differencing value of a pair of pixels belongs to the same level before and after data are hidden. Experimental results have demonstrated that the proposed approaches could obtain a larger embedding capacity and a higher image quality than that of Lee and Chen's method. Furthermore, with the same embedding ratio ER, 4 bpp, the proposed approach could achieve a higher image quality than that of a simple 4-bit LSB. The proposed approaches are the anti-forensics steganography system with the satisfactions of flexibility, adaptability, capacity, and imperceptibility. In this paper, the test, such as statistical steganalyzer is imposed to guarantee the safer evidence behind the image-media. Future works should apply other attack tools to discuss attach-resistant related topics. While only

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TABLE IV
(CONT.)THE RESULTS OF EMBEDDING RATIO (ER) AND PSNRS USING OUR HOEMD APPROACH WITH VARIOUS $< k_1, k_2 >$ VALUES.

Cover image	$< 8, 9 >$		$< 9, 10 >$		$< 10, 11 >$		$< 11, 12 >$		$< 12, 13 >$	
	ER	PSNR	ER	PSNR	ER	PSNR	ER	PSNR	ER	PSNR
Elaine	4	33.78	4	32.82	4	31.98	4.5	31.11	4.5	30.41
Lena	4	33.76	4	32.78	4	31.97	4.5	31.10	4.5	30.23
Baboon	4	33.77	4	32.73	4	31.91	4.5	31.10	4.5	30.21
Peppers	4	33.43	4	32.85	4	31.84	4.5	31.06	4.5	30.20
Toys	4	33.23	4	32.22	4	31.82	4.5	31.01	4.5	30.28
Girl	4	33.68	4	32.70	4	31.96	4.5	31.00	4.5	30.21
Gold	4	33.79	4	32.75	4	31.03	4.5	31.07	4.5	30.18
Barb	4	33.77	4	32.76	4	31.92	4.5	31.13	4.5	30.25
Zelda	4	33.81	4	32.72	4	31.97	4.5	31.15	4.5	30.19
Tiffany	4	33.73	4	32.76	4	31.99	4.5	31.02	4.5	30.22
Average	4	33.68	4	32.71	4	31.84	4.5	31.08	4.5	30.24

TABLE V
THE RESULTS OF EMBEDDING CAPACITY (EC) AND PSNRS USING VARIOUS l-h DIVISIONS WITH DIVIDING LINE D_{lh} = 15.

Cover image	$< 2, 3 >$-$< 7, 8 >$		$< 5, 6 >$- $< 11, 11 >$		$< 8, 8 >$-$< 11, 11 >$	
	EC	PSNR	EC	PSNR	EC	PSNR
Elaine	678,678	41.62	940,912	35.93	1,060,280	33.11
Lena	674,444	41.33	936,588	35.87	1,058,118	33.54
Baboon	717,966	38.66	980,110	33.84	1,079,879	33.10
Peppers	668,006	41.98	930,150	36.63	1,054,899	33.87
Toys	672,900	41.58	935,004	36.78	1,057,346	33.86
Girl	669,670	41.54	931,814	36.59	1,055,731	33.64
Gold	673,366	41.30	935,510	35.88	1,057,579	33.74
Barb	716,788	38.93	978,922	34.99	1,079,285	33.31
Zelda	663,370	42.35	925,510	36.59	1,052,591	34.09
Tiffany	670,086	41.81	932,230	36.47	1,055,939	33.76
Average	680,526	41.11	942,679	35.94	1,061,164	33.76

TABLE VI
COMPARISONS OF THE RESULTS BETWEEN LEE AND CHEN'S APPROACH AND OURS THAT USE AN l-h DIVISION WITH D_{lh} = 15.

Embedding method	Lee and Chen's approach						Our approach					
	n=8			n=16			$< 4, 4 >$-$< 7, 8 >$			$< 8, 8 >$-$< 11, 11 >$		
cover images	ER	IF	PSNR	ER	IF	PSNR	ER	IF	PSNR	ER	IF	PSNR
Elaine	3	0.999534	37.91	4	0.997731	31.41	3.01	0.999575	38.71	4.04	0.998464	33.54
Lena	3	0.999436	37.91	4	0.997233	31.84	3.04	0.999513	38.98	4.04	0.998293	33.10
Baboon	3	0.999461	37.91	4	0.997340	31.86	3.12	0.999361	37.48	4.12	0.998201	33.87
Peppers	3	0.999432	37.92	4	0.997213	31.83	3.02	0.999534	39.29	4.02	0.998401	33.86
Toys	3	0.999188	37.98	4	0.997323	31.86	3.03	0.999499	39.10	4.03	0.998256	33.64
Girl	3	0.999521	37.93	4	0.997641	31.42	3.02	0.999590	39.19	4.03	0.998598	33.74
Gold	3	0.999356	37.93	4	0.996844	31.46	3.03	0.999543	39.04	4.03	0.998131	33.31
Barb	3	0.999479	37.94	4	0.997422	31.86	3.12	0.999372	37.67	4.12	0.998335	34.09
Zelda	3	0.999300	37.93	4	0.996512	31.85	3.02	0.999456	39.85	4.02	0.998133	33.76
Tiffany	3	0.999762	37.93	4	0.998805	31.84	3.03	0.999802	39.21	4.03	0.998309	33.76
Average	3	0.999447	37.93	4	0.997406	31.72	3.04	0.999516	38.85	4.05	0.998412	33.54

introducing one quantization table of the l-h division, the proposed approach can adopt various tables with differencing divisions to identify the exact adjusting strategy.

APPENDIX A

In this section, we used the mathematical theorems to show that our approaches can embed data and extract data correctly, followed by some discussions. In the following theorems, we will show that our HoEMD approach succeeds in embedding and extracting procedures.

Lemma 1. Let $a_i = (2k+1)^i$, for $i = 0, 1, \cdots, n$. Then, $a_i = [2k \times \sum_{j=0}^{i-1}(a_j)] + 1$, for $i = 0, 1, \cdots, n$.

Theorem 1. Let each pixel g_i change among $[k, -k]$. The changed value of pixel g_i is represented as V_i, where $V_i \in [k, -k]$,$i = 0, 1, \cdots, n$. Therefore, there are $(2k+1)^n$ different varieties of the vector $(V_1, V_2, \cdots, V_n)$ for changing n pixels $(g_1, g_2, \cdots, g_n)$.

Theorem 2. Let $f((V_1, V_2, \cdots, V_n)) = a_0V_1 + a_1V_2 + \cdots + a_{n-1}V_n$ is a critical function, where a_{i-1} is a weight and V_i is the changed value of pixel g_i among $[k, -k]$,$i = 0, 1, \cdots, n$. Then, all values of $f = (V_1, V_2, \cdots, V_n)$ mod $(2k+1)^n$ are $(2k+1)^n$ successive integers between 0 and $(2k+1)^{n-1}$.

Proof. We will show that all values of $f = (V_1, V_2, \cdots, V_n)$ are $(2k+1)^n$ successive integers by the following two reasons. Then, we shall have all values of $f = (V_1, V_2, \cdots, V_n)$ mod $(2k+1)^n$ that are $(2k+1)^n$ successive integers between 0 and $(2k+1)^n$-1.

Reason 1: The range of $f = (V_1, V_2, \cdots, V_n)$ is

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TABLE VII
COMPARISONS OF THE RESULTS BETWEEN WU *et al.*'S APPROACH, SIMPLE LSBS WITH 3-BITS AND 4-BITS, AND OURS.

Cover image	PVD and LSB Substution		Simple 3-bit LSB		Simple 4-bit LSB		Our approach			
							$<5,5>$		$<8,8>$	
	ER	PSNR	ER	PSNR	ER	PSNR	ER	PSNR	ER	PSNR
Elaine	2.88	37.25	3	37.88	4	31.39	3	38.14	4	34.31
Lena	2.96	37.61	3	37.49	4	31.22	3	38.16	4	34.38
Baboon	2.75	34.26	3	37.91	4	31.25	3	38.13	4	34.39
Peppers	2.96	37.46	3	37.53	4	31.36	3	37.92	4	34.26
Toys	2.94	36.91	3	37.57	4	31.08	3	37.83	4	34.10
Girl	2.96	37.14	3	37.47	4	31.14	3	38.10	4	34.29
Gold	2.92	37.41	3	37.51	4	31.53	3	38.14	4	34.26
Barb	2.84	36.33	3	37.55	4	31.63	3	38.12	4	34.28
Zelda	2.94	37.61	3	37.53	4	31.14	3	38.15	4	34.41
Tiffany	2.97	37.27	3	37.54	4	31.42	3	38.13	4	34.24
Average	2.91	36.94	3	37.60	4	31.32	3	38.08	4	34.29

$[-\frac{(2k+1)^n-1}{2}, \frac{(2k+1)^n}{2}]$.
The maximum value of $f = (V_1, V_2, \cdots, V_n)$ occurs when $V_i = k$, for $i = 0, 1, \cdots, n$. The maximum value is therefore computed as:

$$\begin{aligned} f = (k, \cdots, k) &= a_0 k + a_1 k + \cdots + a_{n-1} k \\ &= (2k+1)^0 k + \cdots + (2k+1)^{n-1} k \\ &= \frac{(2k+1)^n - 1}{2} \end{aligned}$$

Similarly, the minimum value of $f = (V_1, V_2, \cdots, V_n)$ occurs when $V_i = -k$, for $i = 0, 1, \cdots, n$. The maximum value is therefore computed as:

$$\begin{aligned} f = (-k, \cdots, -k) &= -a_0 k - a_1 k - \cdots - a_{n-1} k \\ &= -(2k+1)^0 k - \cdots - (2k+1)^{n-1} k \\ &= -\frac{(2k+1)^n - 1}{2} \end{aligned}$$

Therefore, the range of $f = (V_1, V_2, \cdots, V_n)$ is $[-\frac{(2k+1)^n-1}{2}, \frac{(2k+1)^n}{2}]$.
Reason 2: Different $(V_1, V_2, \cdots, V_n)$ will result in different is $f = (V_1, V_2, \cdots, V_n)$ values, i.e. function f is a one-to-one function.
Let $(V_1, V_2, \cdots, V_n)$ and $(V_1', V_2', \cdots, V_n')$ not being the same. We will show that $f = (V_1, V_2, \cdots, V_n)$ and $f = (V_1', V_2', \cdots, V_n')$ are not the same. Suppose that m is the largest index such that $V_m \neq V_m'$.
$f = (V_1, V_2, \cdots, V_n)$-$f = (V_1', V_2', \cdots, V_n')$
$= a_0(V_1 - V_1') + a_1(V_2 - V_2') + \cdots + a_{m-1}(V_m - V_m')$ where $V_m - V_m' \in [-2k, -1] \cup [1, 2k]$ and $V_i - V_i' \in [-2k, 2k]$, for $i = 0, 1, \cdots, m-1$.
From Lemma 1, we know that if $(V_k - V_k')$ is positive,
$a_{m-1}(V_m - V_m') \geq a_{m-1}$
$= 2k \times (a_0 + a_1 + \cdots + a_{m-2}) + 1$
$> 2ka_0 + 2ka_1 + \cdots + 2ka_{m-2}$
$\geq a_0(V_1 - V_1') + a_1(V_2 - V_2') + \cdots + a_{m-2}(V_{m-2} - V_{m-2}')$
Therefore, $f(V_1, V_2, \cdots, V_n)$-$f(V_1', V_2', \cdots, V_n') > 0$. Similarly, if $(V_k - V_k')$ is negative, $f(V_1, V_2, \cdots, V_n) - f(V_1', V_2', \cdots, V_n') < 0$.
This concludes that $f(V_1, V_2, \cdots, V_n)$ and $f(V_1', V_2', \cdots, V_n')$ are not the same. ■

Theorem 3. Given a $log_2(\lfloor (2k+1)^n \rfloor)$-length secret $\hat{s}$ and n pixels $(g_1, g_2, \cdots, g_n)$, there is a vector $(V_1, V_2, \cdots, V_n)$ such that $f = ((g_1, g_2, \cdots, g_n) + (V_1, V_2, \cdots, V_n)) \bmod 3^n = \hat{s}$.
Proof.
Let $f(g_1, g_2, \cdots, g_n) \bmod (2k+1)^n$ be equal to F. Therefore,

$$\begin{aligned} & f((g_1, g_2, \cdots, g_n) + (V_1, V_2, \cdots, V_n)) \bmod (2k+1)^n \\ &= \sum_{i=1}^{n} (a_{i-1}(g_i + V_i)) mod (2k+1)^n \\ &= \sum_{i=1}^{n} (a_{i-1} g_i) + \sum_{i=1}^{n} (a_{i-1} V_i) mod (2k+1)^n \\ &= F + f(V_1, V_2, \cdots, V_n) mod (2k+1)^n \end{aligned}$$

From Theorem 2, we know that the found vector $(V_1, V_2, \cdots, V_n)$ is the vector with $f(V_1, V_2, \cdots, V_n) \equiv \hat{s} - F \ mod \ (2k+1)^n$.■

In the following theorems, we will show that our AdEMD approach succeeds in embedding and extracting procedures. First of all, the restrictions of our approach are given. For an l-h division, let ranges R_l and R_h be the ranges of lower-level and higher-level, respectively. In order to apply the l-h division at our approach reasonably and correctly, some restrictions are required for our approach with all theorems shown as follows. The restrictions are $\max\{\lceil \frac{\Psi_1}{2} \rceil, \lceil \frac{\Psi_2}{2} \rceil\} \leq |R_l|$ and $\Psi_1 + \Psi_2 \leq |R_h|$, $|R_l| < |R_h|$.

Definition 1. Let the largest changed value of a pixel be k. The range $[0, k-1]$ is called *Bottom extreme range* and the range $[255-k+1, 255]$ is called *Top extreme range*. Both of the above ranges are called *extreme ranges*.
For the example of $k = 3$, range $[0, 2]$ is *Bottom extreme range* and range $[253, 255]$ is *Top extreme range*.

Definition 2. For a pixel g with the largest changed value k, pixel g is $non-extreme$ if g does not belong to *extreme ranges*.

Lemma 2. Let $< k_1, k_2 >$ be the largest changed values for a pixel-group (g_i, g_{i+1}), S be the secret digit to be embedded into the pixel-group, and (R_1, R_2) be the 2-dimensional vector of S. Suppose that (g_i', g_{i+1}') are the embedded results after Step 3 of the data embedding algorithm of AdEMD is executed. Then, $CF(g_i', g_{i+1}')$ and $CF(g_i, g_{i+1})$ have the same value.
Proof.
From Step 3 of the data embedding algorithm o f AdEMD,

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we have the following equations:

$$\begin{aligned}(V_1, V_2) &= (g_i \ mod\ \Psi_1, g_{i+1} \ mod\ \Psi_2)\\ (R'_1, R'_2) &= (R_1 - V_1, R_2 - V_2)\\ (R''_1, R''_2) &\equiv (g'_i \ mod\ \Psi_1, g'_{i+1} \ mod\ \Psi_2), and\\ (g'_i, g'_{i+1}) &= (g_i + R''_1, g_{i+1} + R''_2).\end{aligned}$$

Therefore,

$$\begin{aligned}g'_i &\equiv g_i + R''_1 \ (mod\ \Psi_1)\\ &\equiv g_i R'_i \ (mod\ \Psi_1)\\ &\equiv g_i + R_1 - V_1 \ (mod\ \Psi_1)\\ &\equiv R_1 \ (mod\ \Psi_1).\end{aligned}$$

Similarly, $g'_{i+1} \equiv R_2 \ (mod\ \Psi_2)$.
Finally, we have

$$\begin{aligned}CF(g'_i, g'_{i+1}) &= (g'_i \ mod\ \Psi_1) + (g'_{i+1} \ mod\ \Psi_2) \times \Psi_1\\ &= (R_1 \ mod\ \Psi_1) + (R_2 \ mod\ \Psi_2) \times \Psi_1\\ &= CF(R_1, R_2). \blacksquare\end{aligned}$$

Lemma 3. Let (g'_i, g'_{i+1}) be the embedded results after Step 3 of the data embedding algorithm of AdEMD is executed. Then, $CF(g'_i, g'_{i+1})$ and $(g'_i \pm z_1\Psi_1, g'_{i+1} \pm z_2\Psi_2)$ have the same value, where z_1 and z_2 are positive integers.
Proof
Because of $(g'_i \pm z_1\Psi_1 \equiv g'_i \ mod\ \Psi_1)$ and $(g'_{i+1} \pm z_2\Psi_2 \equiv g'_{i+1} \ mod\ \Psi_2)$, we have
$CF(g'_i \pm z_1\Psi_1, g'_{i+1} \pm z_2\Psi_2)$
$=(g'_i \pm z_1\Psi_1 \ mod\ \Psi_1)+(g'_{i+1} \pm z_2\Psi_2 \ mod\ \Psi_2) \times \Psi_1$
$=(g'_i \ mod\ \Psi_1) + (g'_{i+1} \ mod\ \Psi_2) \times \Psi_1$
$=CF(g'_i, g'_{i+1})$ $\blacksquare$

Lemma 4. Suppose that two pixels g_i and g_{i+1} are $non-extreme$. Let $< k_1, k_2 >$ be the largest changed values for the pixel-group (g_i, g_{i+1}) and S be the secret digit to be embedded into the pixel-group. Then, $g_i - k_1 \le g'_i \le g_i + k_1$ and $g_{i+1} - k_2 \le g'_{i+1} \le g_{i+1} + k_2$ are held.
Proof.
Let (R_1, R_2) be the 2-dimensional vector of S and $(V_1, V_2) = (g_i \ mod\ \Psi_1, g_{i+1} \ mod\ \Psi_2)$. From Step 3 of the data embedding algorithm of AdEMD, we have the following relations: $0 \le R_j \le 2k_j$ and $0 \le V_j \le 2k_j$, for $j = 1, 2$.
By the fact that $(R'_1, R'_2) = (R_1 - V_1, R_2 - V_2)$, we have $-2k_j \le R'_j \le 2k_j$, for $j = 1, 2$.
For $j = 1, 2$, we show that $R''_j \in [-k_j, k_j]$ by classifying R'_j value into the following three cases:
Case (1): $-2k_j \le R'_j \le -k_j - 1$
In this case, R''_j is set to $R'_j + \Psi_j$. Therefore, $-2k_j + \Psi_j \le R''_j \le -k_j - 1 + \Psi_j$
$\Rightarrow -2k_j + (2k_j + 1) \le R''_j \le -k_j - 1 + (2k_j + 1)$
$\Rightarrow 1 \le R''_j \le k_j$.
Case (2): $-k_j \le R'_j \le k_j$
In this case, R''_j is set to R'_j. Therefore, $-k_j \le R''_j \le k_j$.
Case (3): $k_j + 1 \le R'_j \le 2k_j$
In this case, $R''_j$is set $R'_j - \Psi_j$. Therefore,
$k_j + 1 - \Psi_j \le R''_j \le 2k_j - \Psi_j$
$\Rightarrow k_j + 1 - (2k_j + 1) \le R''_j \le 2k_j - (2k_j + 1)$
$\Rightarrow -k_j \le R''_j \le -1$.
Because of $R''_j \in [-k, k]$, for $j = 1, 2$, and $(g'_i, g'_{i+1}) = (g_i + R''_1, g_{i+1} + R''_2)$, we have $g_i - k_1 \le g'_i \le g_i + k_1$ and $g_{i+1} - k_2 \le g'_{i+1} \le g_{i+1} + k_2$. Since pixel g_i and g_{i+1} are are $non-extreme$, the embedded results (g'_i, g'_{i+1}) are feasible, i.e., the overflow/underflow problems do not happen at pixels g_i and g_{i+1}.$\blacksquare$

Continuing the previous lemma, we have the relation between d_i and d'_i as shown in the following lemma, where $d_i = |g_i - g_{i+1}|$ and $d'_i = |g'_i - g'_{i+1}|$.

Lemma 5. Let d_i be the difference value between g_i and g_{i+1}, and d'_i be the difference value between g'_i and g'_{i+1}. Then, $d_i - k_2 - k_1 \le d'_i \le d_i + k_2 + k + 1$.
Proof
From Lemma 4, we know that $g_i - k_1 \le g'_i \le g_i + k_1$ and $g_{i+1} - k_2 \le g'_{i+1} \le g_{i+1} + k_2$.
Without loss of generality, let $g_i \le g_{i+1}$. Therefore, $d_i = |g_{i+1} - g_i| = g_{i+1} - g_i$. Also, $d'_i = |g'_{i+1} - g'_i|$. We have
$(g_{i+1} - k_2) - (g_i + k_1) \le g'_{i+1} - g'_i \le (g_{i+1} + k_2) - (g_i - k_1)$
$\Rightarrow (g_{i+1} - g_i) - (k_2 + k_1) \le g'_{i+1} - g'_i \le g_{i+1} - g_i) + (k_2 + k_1)$
$\Rightarrow d_i - (k_2 + k_1) \le d'_i \le d_i + (k_2 + k_1)$.$\blacksquare$

Theorem 4. Suppose that both g_i and g_{i+1} are $non-extreme$. With the restrictions $|k_1 + k_2| \le 1$, $\max\{\lceil \frac{\Psi_1}{2} \rceil, \lceil \frac{\Psi_2}{2} \rceil\} \le |R_l|$, and $\Psi_1 + \Psi_2 \le |R_h|$, the readjusting phase works.
Proof.
In the condition that d_i and d'_i fall into different ranges, Fig. 3(a) and (b) show the cases, where d_i belongs to lower-level and higher-level, respectively. From Lemma 5, the maximum difference between d_i and d'_i is k_1+k_2, which has been pointed out in Fig. 4. Because of $|d'_i - d_i| \le k_1 + k_2$, d'_i can go back to the range which d_i belongs to by moving one of g'_i and g'_{i+1} at distances Ψ_1 and Ψ_2, respectively. Now, we show that at least one of g'_i and g'_{i+1} can move. Without loss of generality, let $g'_i \le g'_{i+1}$, and we will discuss the cases of Fig. 4 one by one.
From the restriction $|k_1 - k_2| \le 1$ and the definitions $\Psi_1 = 2k_1 + 1$ and $\Psi_2 = 2k_2 + 1$, we have

$$\begin{aligned}k_1 - 1 \le k_2 \le k_1 + 1 \ &\Rightarrow\ 2k_1 - 1 \le k_1 + k_2 \le 2k_1 + 1\\ &\Rightarrow\ \Psi_1 - 2 \le k_1 + k_2 \le \Psi_1\\ &\Rightarrow\ -2 \le k_1 + k_2 - \Psi_1 \le 0.\end{aligned}$$

Similarly, we have $-2 \le k_1 + k_2 - \Psi_2 \le 0$.
From the restriction $|k_1 - k_2| \le 1$ and the equation $\max\{\lceil \frac{\Psi_1}{2} \rceil, \lceil \frac{\Psi_2}{2} \rceil\}$=$\max\{k_1, k_2\} + 1$, we have
$k_1 + 1 \le \max\{k_1, k_2\} + 1 \le k_1 + 2$
$\Rightarrow k_1 + 1 - \Psi_1 \le \max\{k_1, k_2\} + 1 - \Psi_1 \le k_1 + 2 - \Psi_1$
$\Rightarrow -k_1 \le \max\{k_1, k_2\} + 1 - \Psi_1 \le -k_1 + 1$.
Therefore, we have $-k_1 \le \{\lceil \frac{\Psi_1}{2} \rceil, \lceil \frac{\Psi_2}{2} \rceil\} - \Psi_1 \le -k_1 + 1$, $-k_2 \le \{\lceil \frac{\Psi_1}{2} \rceil, \lceil \frac{\Psi_2}{2} \rceil\} - \Psi_2 \le -k_2 + 1$.
Case of Fig. 3(a):
d_i belongs to lower-level. Therefore, $0 \le d_i \le |R_l| - 1$. Becasue $d'_i \in R_h$, we have $|R_l| \le d'_i \le |R_l| - 1 + k_1 + k_2$. By the restriction $\max\{\lceil \frac{\Psi_1}{2} \rceil, \lceil \frac{\Psi_2}{2} \rceil\} \le |R_l|$, we have $\max\{\lceil \frac{\Psi_1}{2} \rceil \ \lceil \frac{\Psi_2}{2} \rceil\} \le d'_i \le |R_l| - 1 + k_1 + k_2$.
In this case, Fig. 4(a) shows the moving directions of g'_i and g'_{i+1}. At least one of g'_i and g'_{i+1} can move because the sum of the two moving distances $\Psi_1 + \Psi_2$ is less than 256. We have just shown that the case that g'_i can move, which means $g_i + \Psi_1 \le 255$. After g'_i is moved by Ψ_1,

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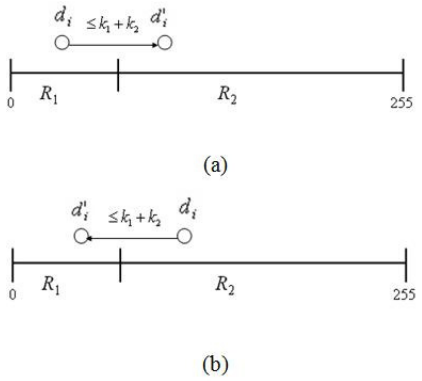

Fig. 3. The cases that d_i and d'_i fall into difference ranges when g_i and g_{i+1} are $non-extreme$: (a) d_i ∈lower-level; (b) d_i ∈higher-level.

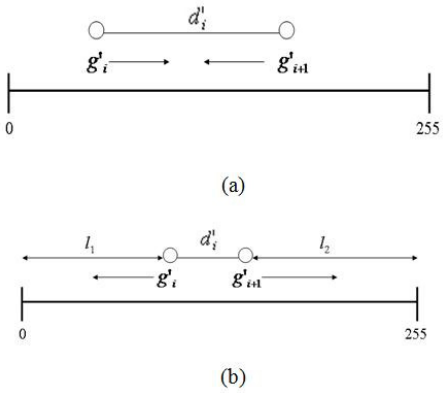

Fig. 4. The corresponding moving operations of g'_i and g'_{i+1} in Fig. 3: (a) d_i ∈lower-level; (b) d_i ∈higher-level.

we have $\max\{\lceil\frac{\Psi_1}{2}\rceil, \lceil\frac{\Psi_2}{2}\rceil\} - \Psi_1 \le g'_{i+1} - (g'_i + \Psi_1) \le |R_l| - 1 + k_1 + k_2 - \Psi_1 \Rightarrow -k_1 \le g'_{i+1} - (g'_i + \Psi_1) \le |R_l| - 1$. Therefore, the new difference $d'_i = |g'_{i+1} - (g'_i + \Psi_1)|$ goes back to the lower-level.

Case of Fig. 3(b):

As shown in Fig. 3(b), the relation between d_i and d'_i is $d_i \le d'_i + (k_1 + k_2)$. Fig. 4(b) shows the moving directions of g'_i and g'_{i+1}. Let l_1 be the width of $[0, g'_i]$ and l_2 be the width of $[g'_{i+1}, 255]$. That is, $l_1 = g'_i - 0$ and $l_2 = 255 - g'_i$. Because $d'_i \in R_l$, we have $d'_i < |R_l|$. So,

$$\begin{aligned} l_1 + l_2 &= 256 - (d'_i + 1) \\ &\ge 256 - |R_l| \\ &= |R_h| \\ &\ge \Psi_1 + \Psi_2 \end{aligned}$$

Therefore, at least one of g'_i and g'_{i+1} can be moved by Ψ_1 and Ψ_2, respectively. We have just shown the case that g'_i can move, which means $g'_i - \Psi_1 \ge 0$. After g'_i is moved by Ψ_1, the new difference is equal to

$$\begin{aligned} d'_i + \Psi_1 &\ge d'_i + (k_1 + k_2) \\ &\ge d_i. \end{aligned}$$

Therefore, the new difference goes back to the higher-level. ■

From Lemma 4, we know that the underflow problem would happen at pixel g if g belongs to $Bottom\ extreme\ range$. Similarly, the overflow problem would happen at pixel g if g belongs to $Top\ extreme\ range$. For the largest changed values $< k_1, k_2 >$ of a pixel-group (g_i, g_{i+1}), the stego-pixel g'_i is adjusted by $\pm\Psi_1$ when the overflow/underflow occurs; the stego-pixel g'_{i+1} is adjusted by $\pm\Psi_2$ when the overflow/underflow occurs. Fig. 5 shows the ranges of pixel g and its stego-pixel g'. For example, if g belongs to $Bottom\ extreme\ range$, we have $0 \le g \le k - 1$ and $0 \le g' \le 2k$. Theorem 4 has shown that the readjusting phase is correct if none of g_i and g_{i+1} belongs to $extreme\ ranges$. Now, we will discuss the cases that at least one of g_i and g_{i+1} belong to $extreme\ ranges$. Without loss of generality, let $g_i \le g_{i+1}$. These cases are divided into three categories: Category A, Category B, and Category C, as follows:

Category A: $g_i, g_{i+1} \in Bottom\ extreme\ range$ or $g_i, g_{i+1} \in Top\ extreme\ range$.

We just showed the case of $g_i, g_{i+1} \in Bottom\ extreme\ range$. The proof of the other case is similar. $g_i, g_{i+1} \in Bottom\ extreme\ range$ means that $0 \le g_i \le k_1 - 1$ and $0 \le g_{i+1} \le k_2 - 1$. Therefore, we have

$$\begin{aligned} d_i = |g_{i+1} - g_i| &\le max\ \{k_1 - 1, k_2 - 1\} \\ &\le max\{\left\lceil\frac{\Psi_1}{2}\right\rceil, \left\lceil\frac{\Psi_2}{2}\right\rceil\}. \end{aligned}$$

That is , d_i belongs to lower-level. Suppose that d'_i belogs to higher-level. Without loss of generality, let $g'_i \le g'_{i+1}$. Therefore, we have $|R_l| \le g'_{i+1} - g'_i \le max\ \{2k_1, 2k_2\}$ $\Rightarrow max\{\lceil\frac{\Psi_1}{2}\rceil, \lceil\frac{\Psi_2}{2}\rceil\} \le g'_{i+1} - g'_i \le max\ \{2k_1, 2k_2\}$. g'_i moves right by Ψ_1. Then, we have $\max\{\lceil\frac{\Psi_1}{2}\rceil, \lceil\frac{\Psi_2}{2}\rceil\} - \Psi_1 \le g'_{i+1} - (g'_i + \Psi_1) \le max\ \{2k_1, 2k_2\} - \Psi_1$ $\Rightarrow -k_1 \le g'_{i+1} - (g'_i + \Psi_1) \le 1$.

Therefore, the new difference $d'_i = |g'_{i+1} - (g'_i + \Psi_1)|$ goes back to the lower-level.

Category B: $g_i \in Bottom\ extreme\ range$ and $g_{i+1} \notin extreme\ ranges$, or $g_i \notin extremeranges$ and $g_{i+1} \in Top\ extreme\ range$.

We just showed the case that g_i belongs to $Bottom\ extreme\ range$ and g_{i+1} is $non-extreme$. The proof of the other case is similar. There are two conditions that d_i and d'_i fall into different ranges.

(i) $d_i \in R_l$ and $d'_i \in R_h$

In this condition, d'_i is larger than d_i. Fig. 6 shows the cases that have the maximum increment from d_i to d'_i. The maximum increment is $k_1 + k_2 + 1$ in Fig. 6(a) and $k_1 + k_2$ in Fig. 6(b). Therefore, we have $d'_i - d_i \le k_1 + k_2$, which is the same as the situation of Fig. 3(a). The same way can be used to show that d'_i can go back to the lower-level by moving one of g'_i and g'_{i+1} at distances Ψ_1 and Ψ_2, respectively.

(ii) $d_i \ln R_h$ and $d'_i \in R_l$

Under this condition, d'_i is smaller than d_i. Fig. 7 shows the case that has the maximum decrement from d_i to d'_i. The maximum decrement is $2k_1 + k_2$. Therefore, we have

$$\begin{aligned} d_i - d_{i+1} &\le 2k_1 + k_2 \\ \Rightarrow d'_i &\ge d_i - (2k_1 + k_2) \end{aligned}$$

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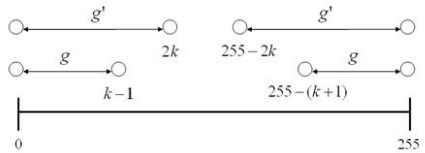

Fig. 5. The ranges of P and P', where belongs to $extreme\ ranges$.

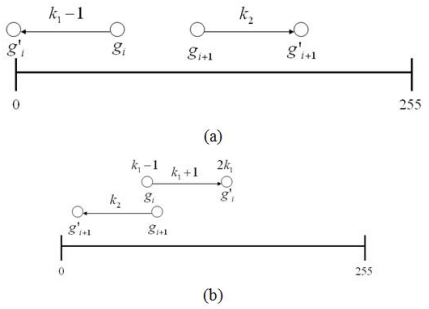

Fig. 6. Two cases that have the maximum increment from d_i to d'_i: (a) Case 1; (b) Case 2.

Under this condition, d'_i may not go back to the higher-level by only moving g'_{i+1} to the right at distance Ψ_2. Hence, the readjusting phase would fail in this case. However, this case can be solved as follows: If d'_i cannot go back to the higher-level by moving g'_{i+1} to the right with distance Ψ_2, one more move of g'_{i+1} is applied.

Category C: $g_i \in Bottom\ extreme\ rang$ and $g_{i+1} \in Top\ extreme\ range$.

Note that $|R_l| + |R_h| = 256$. Therefore, we have

$$\begin{aligned} |R_l| &= 256 - |R_h| \\ &\leq 256 - (\Psi_1 + \Psi_2) \end{aligned}$$

In this category, the maximum value of g_i is $k_1 - 1$ and the minimum value of g_{i+1} is $255 - (k_2 - 1)$ as shown in Fig. 5. Therefore, we have

$$\begin{aligned} d_i = g_{i+1} - g_i &\geq 255 - (k_1 - 1) - (k_2 - 1) \\ &= 256 - k_1 - (k_2 - 1) \\ &\geq 256 - \Psi_1 - \Psi_2 \\ &\geq |R_l|. \end{aligned}$$

Also, the maximum value of g'_i is $2k_1$ and the minimum value of g'_{i+1} is $255 - 2k_2$ as shown in Fig. 5. Therefore, we have

$$\begin{aligned} d'_i = g'_{i+1} - g'_i &\geq 255 - 2k_1 - 2k_2 \\ &= 257 - \Psi_1 - \Psi_2 \\ &\geq |R_l|. \end{aligned}$$

As a result, both d_i and d'_i belong to $|R_h|$.

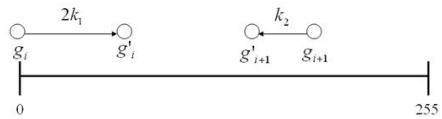

Fig. 7. The case that has the maximum decrement from d_i to d'_i.

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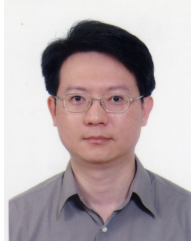

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